

LangChain - Short Notes

What is LangChain?

LangChain is an open-source framework used to build applications powered by Large Language Models (LLMs). It helps developers connect LLMs with external data sources, APIs, databases, and custom workflows.

Key Components

- 1. LLMs:** Language models such as OpenAI, Groq, or Hugging Face.
- 2. Prompts:** Templates that define how instructions are sent to the model.
- 3. Chains:** Sequence of operations combining prompts and models.
- 4. Memory:** Stores conversation history for context-aware responses.
- 5. Retrievers:** Fetch relevant information from documents or vector databases.
- 6. Agents:** Decide which tools to use to solve complex tasks.

Common Use Cases

- Chatbots
- RAG (Retrieval-Augmented Generation)
- Document Question Answering
- AI Assistants
- Multi-Agent Systems
- Workflow Automation

Advantages

- Easy integration with LLMs
- Supports RAG and vector databases
- Modular and scalable architecture
- Works with multiple AI providers
- Simplifies AI application development

Conclusion

LangChain is one of the most popular frameworks for developing modern LLM applications. It enables developers to build intelligent, context-aware, and production-ready AI systems efficiently.